market trends with incoming multi-source financial messages. Additionally, during this phase, FinMem develops a memory module enriched with a comprehensive knowledge base, thereby evolving its capability for independent decision-making in future trading activities.

4 Experiments Setups

We aim to evaluate the trading performance of FinMem. And we further illustrate its unique advantages of requiring significantly less historical trading time window to train and take full use of key financial data time series as well as textual information. Specifically, we conducted several experiments to study the following research questions (**RQs**):

- **RQ1:** Is FinMem capable of outperforming contemporary state-of-the-art algorithmic trading agents?
- **RQ2:** Are there tasks that challenge other trading algorithms but are manageable by FinMem?
- **RQ3:** Which LLM is best suited to form the backbone framework of FinMem?
- **RQ4:** Does equipping FinMem with different risk inclination choices truly differentiate its trading performance?
- **RQ5:** Can FinMem effectively filter and prioritize information to facilitate informed trading decisions?

In the rest of the section, we begin by introducing the real-world financial dataset used in our experiments. We then describe the comparative algorithmic agents and list several widely used financial metrics. Our experiments fall into two categories: 1) The comparative experiments of FinMem versus other algorithmic trading agents, and FinMem using different LLMs as backbone algorithms. 2) The ablation studies evaluate the effects of FinMem's adjustable cognitive span and the role of the trader's dynamic character settings, particularly the risk inclinations, on its trading performance. Through experiments, FinMem demonstrates to outperform other comparative algorithmic agents. Furthermore, we are able to show that its profiling and memory modules are sophisticated and tailored to effectively address the intricacies of the financial landscape, resulting in superior trading performance.

4.1 Datasets And Database Structure:

We assessed FinMem's performance using multi-source financial data from August 15, 2021, to April 25, 2023, sourced from reputable financial databases and APIs like Yahoo Finance (via yfinance) and Alpaca News API, detailed explained in Table 1. The stock tickers used in our comparative experiments are detailed in Figure 4. These were selected because they are among those with the highest volumes of accessible news text data, and they are spread across various trading sectors. This selection provides ample data to evaluate FinMem's generalization capabilities. Additionally, Tesla, Inc. (TSLA) was specifically chosen for ablation studies due to its association with the largest amount of textual data, offering sufficient information to assess performance differences for the FinMem's key features like cognitive spans.

The raw multi-source input data, initially stored in the "Raw Financial Data Warehouse", are diverged into FinMem's "Layered Long-term Memory Data Warehouse" based on timeliness through working memory's summarization operation in Figure 2. The deep processing layer holds annual reports (Form 10K's) insights, the intermediate layer contains quarterly reports (Form 10Q's) insights, and the shallow layer accommodates daily financial news insights.

We leveraged the open-source vector database FAISS [23] for constructing the memory warehouse of FinMem, benefiting from its rapid querying in high-dimensional vectors and compatibility with OpenAI for cosine-similarity-based semantic searches on specific tickers. This setup facilitates efficient top-ranked event retrieval. Data categorization and memory module workflow are also illustrated in Figure 2

4.2 Baseline And Comparative models:

We assess FinMem's trading performance in comparison to five advanced algorithmic agents and a commonly accepted baseline trading strategy. Among these, three models employ Deep Reinforcement Learning (DRL) approaches, while the remaining two are based on Large Language LLMs. Brief descriptions of each are provided below:

Buy-and-Hold strategy (B&H):

A passive investment approach, where an investor purchases stocks and holds onto them for an extended period regardless of market fluctuations, is commonly used as a baseline for comparison of stock trading strategies.

DRL trading agents:

Raw Financial Data Warehouse
News data associated with ticker indexes: News data is sourced from the Alpaca News API, which utilizes Benzinga as its backend provider.
Corporate quarter filings indexes: Quarterly reports (Form 10-Q) are required by the U.S. Securities and Exchange Commission (SEC).
Corporate annual filings indexes: Annual reports (Form 10-K) required by the U.S. Securities and Exchange Commission (SEC).
Stock price records: Daily stock open-high-close-volume (OHLCV) data from Yahoo Finance.

FINMEM's Layered Long-term Memory Data Warehouse
Shallow Processing: Insights of real-time market news extracted by LLM. Updated daily.
Intermediate Processing: Insights of 10-Q filings extracted by LLM. Updated quarterly.
Deep Processing: 10-K fillings, all summarized by LLM. Updated yearly. Extended reflections for the stock in response to the trading inquiry include FINMEM's cumulative trading returns, decision-making processes, trade volumes, and underlying reasons. Updated daily.

Table 1: Raw data and memory warehouses of FINMEM

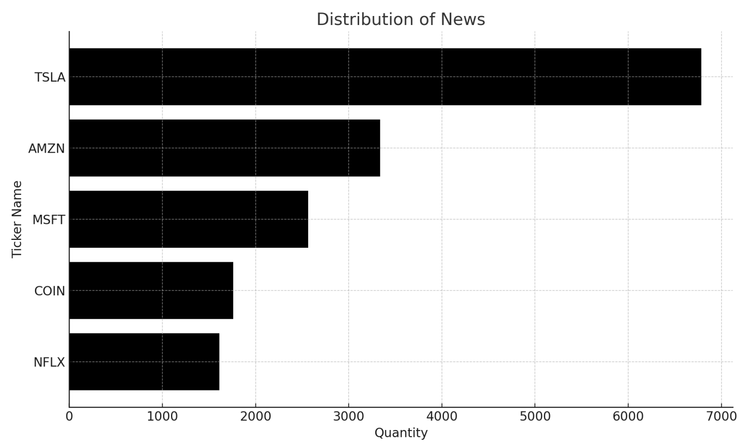

Figure 4: The distribution of news in scraped from Alpaca News API for the five stocks in the experiments

As the FINMEM is practiced and examined on the basis of single stock trading and discrete trading actions, we choose three advanced DRL algorithms fitting into the same scenarios according to the previous and shown expressive performance in the work of Liu et al. [28, 26]. The DRL training agents only take numeric features as inputs.

- **Proximal Policy Optimization (PPO):** PPO [42] is employed in stock trading due to its stability and efficiency. One salient advantage of PPO is that it maintains a balance between exploration and exploitation by bounding the policy update, preventing drastic policy changes.
- **Deep Q-Network (DQN):** DQN [32] is an adaptation of Q-learning, that can be used to optimize investment strategies. Unlike traditional Q-learning that relies on a tabular approach for storing Q-values, DQN generalizes Q-value estimation across states using deep learning, making it more scalable for complex trading environments.
- **Advantage Actor-Critic (A2C):** A2C [31] is applied to optimize trading actions in the financial environment. It operates by simultaneously updating both the policy (actor) and the value (critic) functions, providing a balance between exploration and exploitation.

LLM trading agents:

We evaluate FINMEM against two LLM agents in the context of stock trading. The first LLM agent, known for its proficiency in general-purpose tasks, serves as a baseline. The second agent, a leading-edge LLM in trading, has been acclaimed for its promising performance in stock market operations.

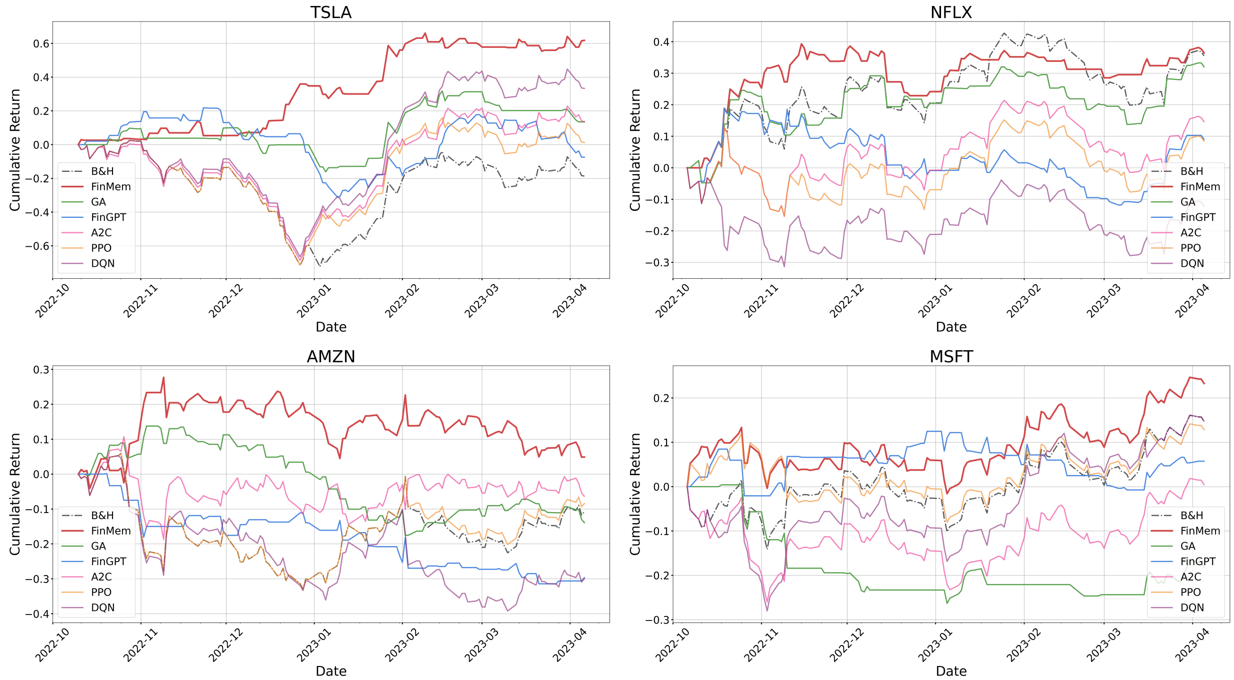

Figure 5: Cumulative return comparison over time between FINMEM and other algorithmic agents across five stocks.

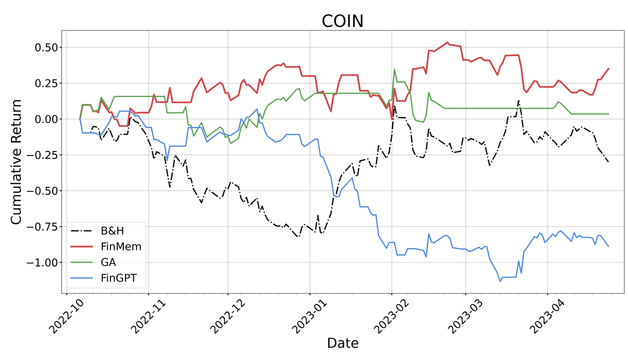

Figure 6: Cumulative Return comparison over time between FINMEM and other algorithmic agents on Coinbase Global, Inc. (COIN).

- **General-purpose Generative Agents – GA:** The generative AI agent by Park et al. [36], originally intended to simulate realistic human behavior and make everyday decisions, has been adapted here for specific stock trading tasks. This agent's architecture includes a memory module that employs recency, relevance, and importance metrics to extract pivotal memory events for informed decision-making. However, it does not provide a layered memory module to effectively differentiate the time sensitivities unique to various types of financial data. Additionally, although it features a profiling module to define agent attributes like professional background, the model does not include a mechanism for self-adaptive risk preference. In our experiments, we modified the original prompt template created by Park et al., which was intended for general daily tasks, to suit financial investment tasks. The textual elements of this revised template closely align with those of FINMEM, with the exception of two components that are absent in this version of general-purpose Generative Agents.

- **LLM trading agents – FINGPT:** A novel open-source LLM framework specialized for converting incoming textual and numeric information into informed financial decision-making, introduced by Yang et al. [55]. It claims superiority over the traditional buy-and-hold strategy.